

The Medial Antiprism: Two Strands, a Horizon at One Half and an Undecided Trefoil

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Abstract

Stack triangles along an axis, each turned through 180° against the one below, and join every vertex of each triangle to every side midpoint of its neighbours. The side midpoints of a triangle are its vertices multiplied by $-\frac{1}{2}$, so one move, a half-turn with a halving, builds the whole object, and consecutive layers form triangular antiprisms. We call it the medial antiprism and ask what survives when each step passes only a sum. With six layers, its 36 nodes and 90 lines form two strands that share every layer and no line, exchanged by the central inversion. Between layers each strand applies $J = I + C + C^2$, which keeps the still pattern and erases the two patterns that turn by a third. In every gap each strand crosses itself three times, at one third and at two thirds of the height, and the two strands meet six times at one half; moving the midpoints to the fraction f of their sides moves the meetings to f and $1 - f$. Each bridge is a Möbius ladder whose rim is a six-edge trefoil with its three crossings undecided, and a ring of an odd number of layers joins the two strands into one that returns only after two circuits. Unrolled, the routes of a strand form the 3-adic tree, whose ends are the 3-adic integers written in the digits $\{-1, 0, 1\}$; in the tree's variable s the still pattern of a bridge sits at $s = 1$ and its turning patterns on $\Re s = \frac{1}{2}$. Read along the axis, a bridge is a transfer matrix that kills the turning patterns in one step, and with triad spins on the nodes the chain has a gap at every coupling and its Lee–Yang zeros on the unit circle.

Keywords: medial triangle; triangular antiprism; complete bipartite graph; Möbius ladder; trefoil knot; Ihara zeta function; Ramanujan graph; Bruhat–Tits tree; transfer matrix; Lee–Yang theorem

1. Introduction

The side midpoints of a triangle form its medial triangle, which is the triangle itself turned through 180° about its centre and halved: if the vertices are v_0, v_1, v_2 with $v_0 + v_1 + v_2 = 0$, the midpoint of the side opposite v_j is $-\frac{1}{2}v_j$. Turn the triangle through 180° and double it, and its vertices land on the directions of the old midpoints. This paper studies the object that one move, a half-turn with a halving, builds when it is repeated along an axis. Layers alternate between a triangle and its half-turned copy, so each pair of neighbouring layers is a triangular antiprism. Every vertex of a layer is joined to every side midpoint of the next. We call the result the *medial antiprism* (Definition 1).

One question organises the paper: *what survives when each step passes only the sum?* Between neighbouring layers the lines join three nodes completely to three, so what passes is the sum of three values (Proposition 3). The sum, the still pattern, survives every step, and the two patterns that turn by a third are erased at the first. The geometry records where the erased parts meet. In every gap each strand meets itself at one third and two thirds of the height, and the two strands meet each other at one half (Theorem 5).

The same object appears in several forms, and Figure 1 shows them side by side. It is a solid built of lines (panel a), generated by one move (b), made of bridges that are Möbius ladders (c), split into two strands (d), cut by planes at the thirds and the half (e), acting as a filter on patterns (f), unrolled into a tree (g), and smoothed into a standing wave (h). Sections 2–7 work in space. From Section 8 on,

the positions of the nodes are forgotten and only the lines are used. Each section names the form it uses.

Several facts are immediate once the coupling J is seen. These are the Kronecker form of the spectrum, the Ramanujan property of the bridge, and the Lee–Yang circle, which follows from Newman’s theorem; they are stated as such. The geometric results carry the weight of the paper: the placement law of the meetings (Theorem 3), the slab of the thirds and the half (Theorem 5), the undecided trefoil (Theorem 4), the odd ring (Proposition 1), and the unrolled strand (Theorem 8). We build on the spectral theory of graphs [4], on Fortescue’s symmetrical components [6], on Ihara’s zeta function and its Ramanujan form [3,13,19,25], on Guy and Harary’s Möbius ladders [10] and Randell’s stick number [22], on the Bruhat–Tits tree [24] and its use in p -adic holography [8], and on the theorems of Lee and Yang, Griffiths and Newman [9,17,21]. Each is cited where it is used.

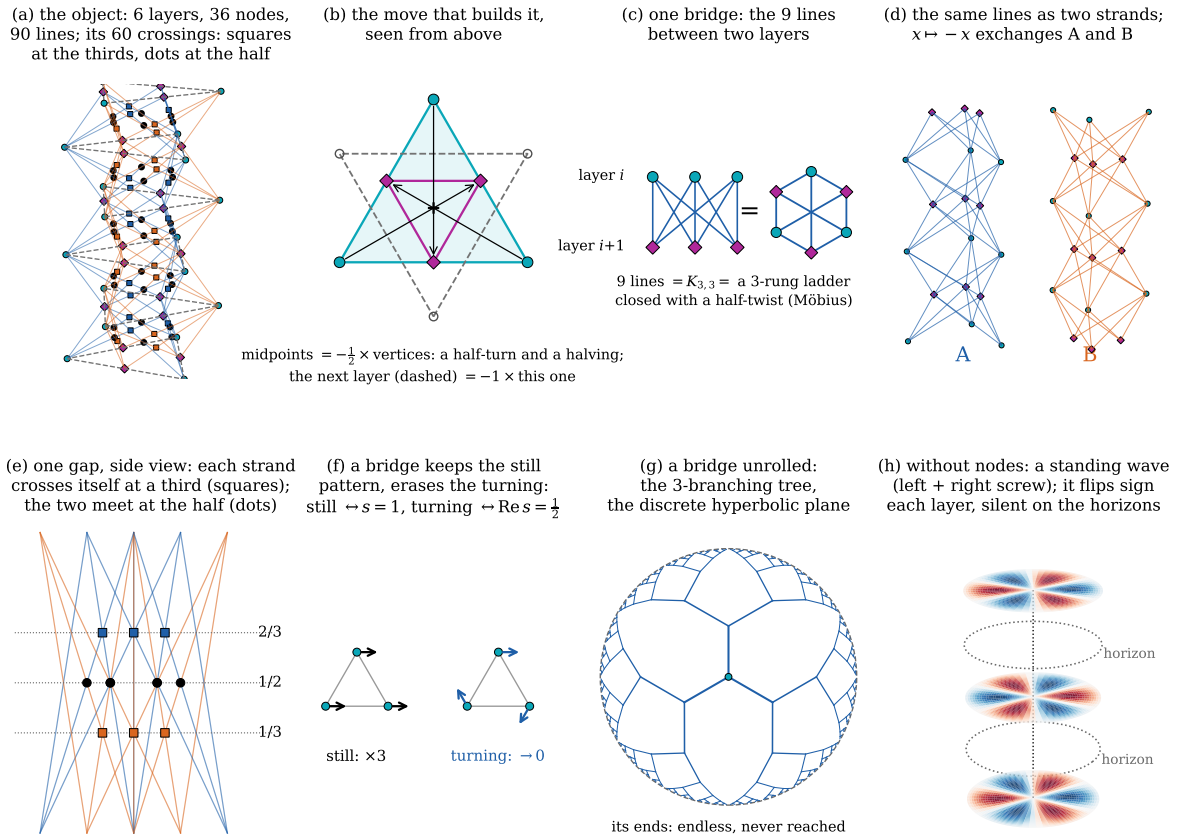


Figure 1. One object in eight forms. (a) The medial antiprism with six layers: cyan vertices, magenta midpoints, the 90 lines coloured by strand, the 60 crossings (squares at the thirds, dots at the half). (b) The generating move from above: midpoints are $-\frac{1}{2}$ times the vertices, and the next layer is the half-turned triangle (Section 2). (c) One bridge, $K_{3,3}$, is the Möbius ladder (Section 6). (d) The two strands, exchanged by the inversion (Section 3). (e) One gap from the side (Section 7). (f) What a bridge keeps and erases (Sections 8 and 10). (g) A bridge unrolled (Section 11). (h) Without nodes (Section 13).

2. Construction

Definition 1 (the medial antiprism). Fix $R > 0$, $Y > 0$ and $L \geq 2$, and let $u_\theta = (\cos \theta, \sin \theta)$ in the horizontal plane. For $i = 1, \dots, L$ put $\theta_i = \frac{\pi}{2}$ for i odd and $\theta_i = -\frac{\pi}{2}$ for i even, and let layer i lie at height $y_i = (i - \frac{L+1}{2})Y$. Its vertices are

$$v_{i,j} = (R u_{\theta_i + 2\pi j/3}, y_i), \quad j = 0, 1, 2,$$

and its midpoints are $m_{i,j} = \frac{1}{2}(v_{i,j+1} + v_{i,j+2})$, indices modulo 3. The lines are the segments $[v_{i,j}, m_{i\pm 1,k}]$ for all j, k and all pairs of neighbouring layers. The configuration is the medial antiprism $\mathcal{A}_L$. Unless stated otherwise $L = 6$, $R = 6$ and $Y = \frac{9}{2}$.

Horizontally $m_{i,j} = -\frac{1}{2}v_{i,j}$, at radius $\frac{R}{2}$ opposite to $v_{i,j}$, and the vertices of layer $i + 1$ are -1 times those of layer i . Seen from above, two neighbouring layers form a six-pointed star (Figure 1b). We call vertices *cyan* and midpoints *magenta*. The nine lines between the vertices of one layer and the midpoints of the next form a *bridge*, and each gap between layers holds two bridges.

3. Two Strands

Form: the lines as joins of their endpoints. Let A be the adjacency matrix of the lines on the $6L$ nodes. Let $\Gamma_c = +1$ on vertices and -1 on midpoints, and $\Gamma_\ell = (-1)^{i+1}$ on layer i .

Theorem 1 (two strands). (i) $A\Gamma_c = -\Gamma_c A$ and $A\Gamma_\ell = -\Gamma_\ell A$; hence A commutes with $\Gamma = \Gamma_c \Gamma_\ell$.
(ii) The lines have exactly two connected components: the strand $A = \{\Gamma = +1\}$, made of the vertices of odd layers and the midpoints of even layers, and the strand $B = \{\Gamma = -1\}$. Each has $3L$ nodes and one level of three nodes in every layer.

Proof. (i) A line joins a vertex of layer i to a midpoint of layer $i \pm 1$, reversing both signs. (ii) Γ is constant along lines, so components lie in its two classes, and within a class consecutive levels are completely joined. $\square$

With its nodes ordered level by level, a strand is the path graph P_L blown up threefold: its adjacency matrix is $A(P_L) \otimes J$, where J is the 3×3 matrix of ones.

Remark 1 (where the strands touch in space). *Theorem 1 concerns lines as joins of their endpoints. In space the 90 segments are not disjoint. Every midpoint $m_{i,j}$ lies on the side $[v_{i,j+1}, v_{i,j+2}]$ of a triangle of the other strand, and lines meet at interior points (Theorem 5). What keeps the strands apart is the conservation of Γ , the product of the two gradings.*

Proposition 1 (the ring). *Joining layer L to layer 1 by the same rule, regarded combinatorially, gives two strands when L is even and one when L is odd. For L odd the single strand passes through every layer twice and returns to its starting node only after two circuits: the strands form the connected double cover of the ring of layers.*

Proof. For L even the layer parity is defined around the ring and Γ is conserved. For L odd the closing bridge joins layers of the same parity, reversing the colour and keeping the parity, so it carries each strand into the other; following a strand once around the ring changes Γ , and twice restores it. $\square$

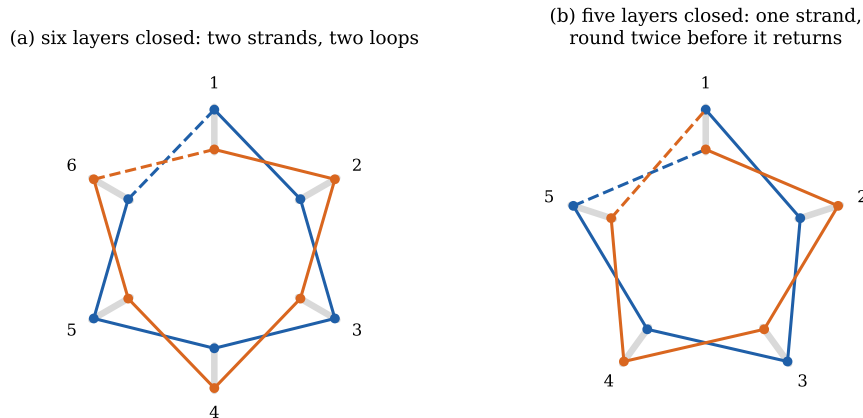


Figure 2. The ring of Proposition 1, one dot per level and the closing bridge dashed. (a) Six layers: two strands, two loops. (b) Five layers: one strand, which comes back to its start only after two circuits.

An odd ring is finite, and the path along its single strand never ends: it returns to the same node only after it has passed every layer twice.

4. Symmetry

Form: the solid in space.

Theorem 2 (symmetry). *The isometries of $\mathbb{R}^3$ that map nodes to nodes of the same colour and lines to lines form the point group D_{3d} of order 12: the identity, rotations by $\pm 120^\circ$ about the axis, three reflections in vertical planes, the inversion through the centre, two rotary reflections by $\pm 60^\circ$ and three rotations by 180° about horizontal axes.*

- (i) *The six elements that keep the vertical order fix each strand; the six that reverse it exchange the strands.*
- (ii) *Each strand is fixed by three reflections, and each strand and the whole are achiral.*
- (iii) *The nodes of one strand lie in the vertical half-planes at $90^\circ, 210^\circ, 330^\circ$ and those of the other at $30^\circ, 150^\circ, 270^\circ$; along each half-plane the radius alternates between R and $\frac{R}{2}$, in opposite phase for the two strands.*

Proof. A symmetry preserves the set of layer planes and the axis, so it keeps or reverses the order of the layers. The order-keeping symmetries fix every layer, whose coloured configuration has symmetry group D_3 ; consecutive layers share its axes and planes, so the order-keeping subgroup is C_{3v} . The inversion maps layer i to layer $L + 1 - i$ turned through 180° and is a symmetry; the group is $C_{3v} \times \{1, \iota\} = D_{3d}$. (i) Order-keeping elements keep both gradings and hence Γ ; order-reversing elements change the parity of the layer and keep the colour, reversing Γ . (ii) and (iii) follow. $\square$

Along each half-plane the radius goes $R, \frac{R}{2}, R, \frac{R}{2}, \dots$: every halving is followed by a doubling. A route along a strand may keep its angle or turn by $\pm 120^\circ$ at each step. Of the $3^6 = 729$ top-to-bottom routes, 3 keep their angle, 3 always turn by $+120^\circ$, 3 always by -120° , and 720 are mixed. Turning a triangle through 180° is turning it through $\pm 60^\circ$, so the stack prescribes no sense of rotation. Handedness first appears in the two trefoils of Section 6, between which the stack does not decide.

5. The Horizon

Form: one gap in space. To see what fixes the place where the strands meet, we move the magenta nodes along their sides. For $f \in (0, 1)$ let $\mathcal{A}_L^{(f)}$ be the stack in which the magenta node of the side from $v_{i,j}$ to $v_{i,j+1}$ is $v_{i,j} + f(v_{i,j+1} - v_{i,j})$, so that $\mathcal{A}_L^{(1/2)} = \mathcal{A}_L$. Heights inside a gap are measured as fractions of Y from the lower layer.

Theorem 3 (the placement law). *For every $f \in (0, 1)$, in each gap of $\mathcal{A}_L^{(f)}$ a line of one strand and a line of the other meet at an interior point in exactly six cases: three at height f and three at height $1 - f$. Each meeting joins the line ending at a magenta node m' of the upper layer to the line starting at the magenta node m of the lower layer with $m' = -m$ horizontally. For $f = \frac{1}{2}$ all six lie on the middle plane, at radius $\frac{\sqrt{3}}{4}R$.*

Proof. Put the lower layer at height 0 with vertices Ru_1, Ru_2, Ru_3 , where $u_1 + u_2 + u_3 = 0$ and any two u_j are independent. The upper layer is turned through 180° , so its vertices are $-Ru_j$ and its magenta nodes are the negatives of the lower ones. A line of one strand runs from Ru_a to $-m_c$, a line of the other from $m_{c'}$ to $-Ru_b$, with $m_c = R((1 - f)u_c + fu_{c+1})$; at height h they are at

$$R[(1 - h)u_a - h(1 - f)u_c - hf u_{c+1}], \quad R[(1 - h)(1 - f)u_{c'} + (1 - h)f u_{c'+1} - h u_b].$$

The only relation among u_1, u_2, u_3 is their sum, so the points coincide exactly when the coefficients of their difference are equal: two linear equations in h with coefficients polynomial in f . Over the 81 choices of (a, c, c', b) there are exactly six solutions valid for every f , all with $c = c'$, three with $h = f$ and three with $h = 1 - f$; the determinants of the remaining systems admit no further solution with $f \in (0, 1)$ and $0 < h < 1$. For $f = \frac{1}{2}$ every meeting point has the form $R(\frac{1}{2}u_a + \frac{1}{4}u_b)$ with $a \neq b$, of radius $\frac{R}{4}|2u_a + u_b| = \frac{\sqrt{3}}{4}R$. $\square$

The meetings come in the mirror pair of heights f and $1 - f$, exchanged by the reversal of the gap. They coincide on the middle plane exactly at the midpoint placement $f = \frac{1}{2}$, the placement of the construction. We call the middle plane of each gap its *horizon*: the only place where the two strands meet when the meetings of their mirror images are to fall there too. The same law holds for stacks of regular pentagons and heptagons (Figure 3b).

Proposition 2 (radial placement). *With the magenta nodes at radius ρR in the same directions, lines of the two strands meet if and only if $\rho \in \{\frac{1}{2}, 1, 2\}$: at height $\frac{1}{2}$ for $\rho = \frac{1}{2}$ and $\rho = 2$, and at heights $\frac{1}{3}$ and $\frac{2}{3}$ for $\rho = 1$.*

Proof. The same comparison with h and ρ unknown has exactly four solutions,

$$(h, \rho) \in \{(\frac{1}{2}, \frac{1}{2}), (\frac{1}{2}, 2), (\frac{1}{3}, 1), (\frac{2}{3}, 1)\},$$

attained by 24 choices of indices. $\square$

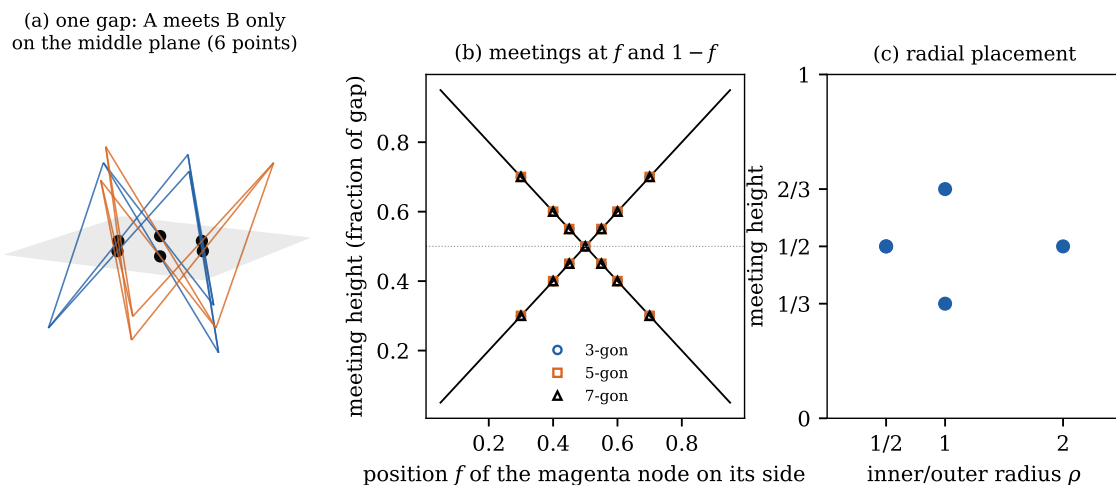


Figure 3. (a) One gap of $\mathcal{A}_6$: the lines of the two strands meet at six points, all on the horizon. (b) Meeting heights against the position f of the magenta nodes, computed for triangles, pentagons and heptagons, with the lines $h = f$ and $h = 1 - f$. (c) Radial placement: meetings only at $\rho = \frac{1}{2}, 1, 2$.

6. The Bridge Loop

Form: one bridge in space, then as a band. In a bridge from the vertices v_a of one layer to the magenta nodes m'_b of the next, the line $[v_a, m'_a]$ lies in a vertical plane through the axis; we call these three lines *radial* and the other six *twisted*. The Möbius ladder M_{2n} is the cycle C_{2n} with the n chords joining opposite vertices [10], and $M_6 \cong K_{3,3}$.

- Theorem 4** (the bridge loop). (i) *The six twisted lines form one hexagon $H = (v_1, m'_2, v_3, m'_1, v_2, m'_3)$, and the three radial lines join opposite vertices of H : the bridge is the Möbius ladder M_6 with rim H .*
- (ii) *H turns by $+120^\circ$ about the axis at each step, so its winding number about the axis is 2; it rises and falls three times.*
- (iii) *The vertical projection of H has exactly three double points, and at each the two segments meet in space, at the point $\frac{2}{3}$ of the way from the vertex end of both, at radius $\frac{R}{3}$.*
- (iv) *Separating H at the three meetings with crossing signs $\varepsilon_1, \varepsilon_2, \varepsilon_3 \in \{\pm 1\}$ gives the closure of the two-strand braid σ_1^n with $n = \varepsilon_1 + \varepsilon_2 + \varepsilon_3$: the right-handed trefoil for $n = 3$, the left-handed trefoil for $n = -3$, and the unknot for $n = \pm 1$, the six mixed cases.*

Proof. (i) The twisted lines are the complement in $K_{3,3}$ of the perfect matching of radial lines, a 6-cycle; each radial line joins vertices at distance 3 on it. (ii) m'_b lies above the direction of v_b , and the angular index increases by one at each step. (iii) The point at parameter $\frac{2}{3}$ on $[v_a, m'_c]$ has horizontal position $\frac{1}{3}Ru_a + \frac{2}{3} \cdot \frac{R}{2}u_c = \frac{R}{3}(u_a + u_c)$ and height $\frac{2}{3}$, an expression symmetric in a and c , so it lies on $[v_c, m'_a]$ too; a direct computation shows no other double point in the projection. (iv) No segment of H passes over the axis, so the angle about the axis increases monotonically along H , through two full turns: the projection is the closure of a braid on two strands with three crossings [1]. A braid on two strands is a power of the single generator σ_1 , and $\sigma_1^{\varepsilon_1}\sigma_1^{\varepsilon_2}\sigma_1^{\varepsilon_3} = \sigma_1^n$. The closure of $\sigma_1^{\pm 3}$ is a trefoil and that of $\sigma_1^{\pm 1}$ is the unknot. The Alexander polynomials [2], $t^2 - t + 1$ and 1, were confirmed by computation. $\square$

No polygon with five or fewer edges is knotted, and six edges are the fewest that make a trefoil [22]. The bridge holds such a six-edge polygon between the two trefoils. Its three crossings are undecided, and a single consistent decision at all three gives a knot of one hand or the other.

The braid has a smooth form. The closure of σ_1^n is the rim of a band with n half-twists. One half-twist is the Möbius band, and three are the band whose rim is a trefoil. In Figure 5a the half-twist is concentrated at the midpoint of the band, as the turn of the triangle happens between neighbouring layers. Panel (b) draws the ladder $M_6 \cong K_{3,3}$ on it, with three rungs joining rim nodes of opposite colour. The bridge is thus a band whose number of half-twists, ± 1 or ± 3 , is left undecided (Figure 6).

(a) the six twisted lines: one loop through itself at 3 points; dotted: 3 rungs

(b) top view: the trefoil shadow, winding number 2, three double points

(c) resolving the three points

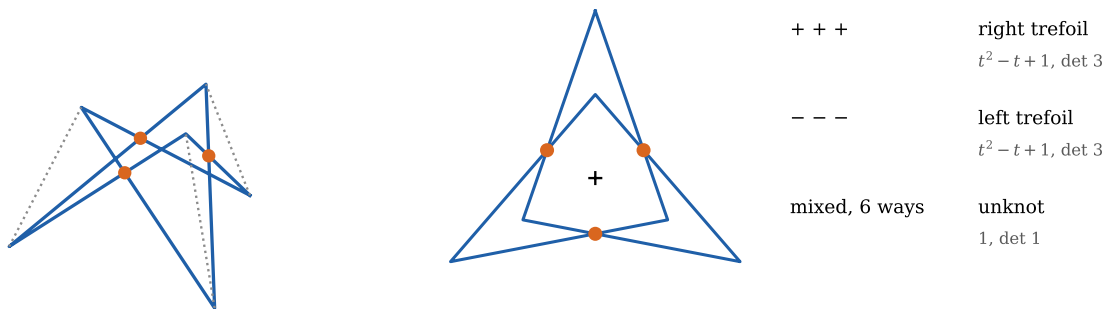


Figure 4. (a) The six twisted lines of one bridge form one hexagon, meeting itself at three points (orange); the radial lines (dotted) are the rungs of the Möbius ladder. (b) The vertical projection: winding number 2, three double points. (c) The resolutions (Theorem 4(iv)).

(a) a Möbius band: flat on both halves, turned through 180° at the midpoint

(b) its rim is one closed curve; three rungs across make the Möbius ladder $M_6 \cong K_{3,3}$, the bridge

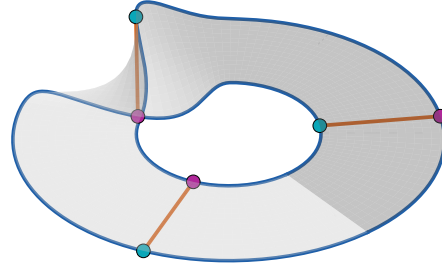
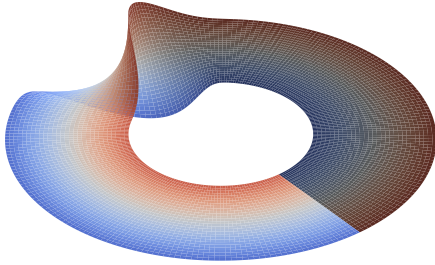


Figure 5. (a) A band with one half-twist, flat on both halves and turned through 180° at its midpoint. (b) Its rim is one closed curve; three rungs across it, joining rim nodes of opposite colour, make the Möbius ladder $M_6 \cong K_{3,3}$.

(a) a band with three half-twists

(b) its single rim is a trefoil

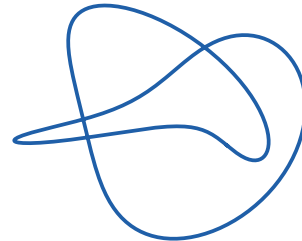
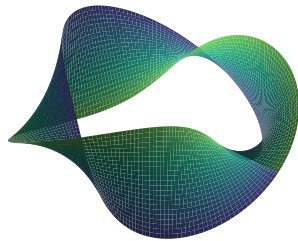


Figure 6. (a) A band with three half-twists. (b) Its rim, a trefoil: the bridge loop with its three crossings decided alike.

7. The Slab of the Thirds and the Half

Form: one gap in space, all crossings at once.

Theorem 5 (the slab). *In each gap of $\mathcal{A}_L$ the eighteen lines cross at exactly twelve interior points. The strand whose vertices lie in the lower layer crosses itself three times at height $\frac{2}{3}$; the other strand crosses itself three times at height $\frac{1}{3}$; the two strands meet six times at height $\frac{1}{2}$. The six self-crossings lie at radius $\frac{R}{3}$ and the six meetings at radius $\frac{\sqrt{3}}{4}R$. From one gap to the next the two strands exchange the heights $\frac{1}{3}$ and $\frac{2}{3}$.*

Proof. Each gap holds one bridge of each strand. The bridge of the strand with vertices below runs from height 0 at its vertex end to height 1, so by Theorem 4(iii) it crosses itself at height $\frac{2}{3}$. The other bridge has its vertex end at height 1, so its crossings lie $\frac{2}{3}$ of the way down, at height $\frac{1}{3}$. The meetings of the two strands are those of Theorem 3 with $f = \frac{1}{2}$. All crossings of all pairs of segments were computed (Section 14), and there are no others. In the next gap the layer parities are exchanged, and so are the strands. $\square$

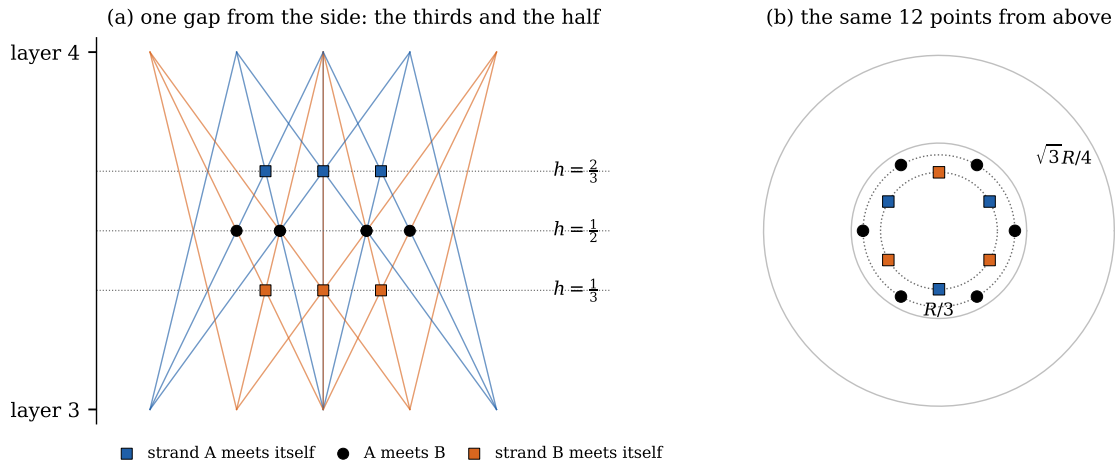


Figure 7. (a) One gap of $\mathcal{A}_6$ from the side: strand A meets itself at $\frac{2}{3}$, strand B at $\frac{1}{3}$, and the two meet at $\frac{1}{2}$ (Theorem 5). (b) The same twelve points from above, on the circles of radius $\frac{R}{3}$ and $\frac{\sqrt{3}}{4}R$.

In every gap, then, a strand meets itself only at the thirds and meets the other strand only at the half. The radial placement of Proposition 2 moves the meetings of the two strands from the half to the thirds.

8. The Triad Between Layers

From here on the positions of the nodes are forgotten; only the lines are used. Order the three nodes of each level by angle. Consecutive levels of a strand are joined by all nine lines, so the coupling is J . Let C be the cyclic shift and $\omega = e^{2\pi i/3}$.

- Proposition 3** (the bridge is the triad). (i) $J = I + C + C^2$. Its eigenvectors are the still pattern $\mathbf{1} = (1, 1, 1)$ and the turning patterns $\tau_{\pm} = (1, \omega^{\pm 1}, \omega^{\pm 2})$, with eigenvalues $1 + z + z^2$ at $z = 1, \omega, \omega^2$: $3, 0, 0$.
- (ii) $J^k = 3^{k-1}J$. The map from the top level of a strand to its bottom level is $3^{L-2}J$, of rank one, and each top node reaches each bottom node by 3^{L-2} walks (81 for $L = 6$).
- (iii) If $x = s\mathbf{1} + n$ with n of mean 0 and covariance $\sigma^2 I$, then $Jx = (3s + \langle n, \mathbf{1} \rangle)\mathbf{1}$: the zero-sum part of n is erased, the mean part is multiplied by 3, and the ratio of signal amplitude to noise amplitude grows by $\sqrt{3}$.

Proof. (i) $I + C + C^2$ has every entry 1; the eigenvalues of C are $1, \omega, \omega^2$ and $1 + \omega + \omega^2 = 0$. (ii) $J^2 = 3J$, and walks from the top level to the bottom one advance one level per step. (iii) $Jn = \langle n, \mathbf{1} \rangle \mathbf{1}$, and $\langle n, \mathbf{1} \rangle$ has variance $3\sigma^2$. $\square$

A bridge passes one number per level, the sum, so the turning patterns are erased once, at the first bridge, and nothing further is lost. In terms of the turn C by a third about the axis, the still pattern has angular momentum 0 modulo 3 and the turning patterns ± 1 : a complete join carries no angular momentum. On the unit circle the generating function of the triad is

$$z^{-1} + 1 + z = 1 + 2\cos\theta = \chi_1(\theta),$$

the character of the spin-one representation of $SU(2)$ at rotation angle θ [23]. It vanishes exactly at $\theta = \pm \frac{2\pi}{3}$, one third and two thirds of a turn.

The same algebra appears in two familiar places. In polyphase circuits $\frac{1}{3}J$ is the projection onto Fortescue's zero-sequence component, and τ_+ and τ_- are the positive and negative sequences [6]. A balanced three-phase system, such as the currents of Tesla's polyphase motor [26], lies entirely in the turning patterns, and the bridge passes nothing of it, as a balanced three-phase load needs no

return wire. In nuclear magnetic resonance a spin- $\frac{1}{2}$ nucleus such as ^{13}C , coupled with constant $\mathcal{J}$ to a spin-one deuteron whose three states are equally populated, has its coherence modulated by $\frac{1}{3}\chi_1(2\pi\mathcal{J}t)$: the 1:1:1 triplet. The modulation vanishes exactly at $t = 1/(3\mathcal{J})$ and $t = 2/(3\mathcal{J})$ [18].

Proposition 4 (the reading of three clocks). *If the three nodes of a level carry the phases $1, e^{i\nu t}, e^{2i\nu t}$, the next node reads $|1 + e^{i\nu t} + e^{2i\nu t}| = |\chi_1(\nu t)|$, which equals 3 at $t = 0$, 2 at $\frac{1}{6}$, 1 at $\frac{1}{2}$ and vanishes exactly at $\frac{1}{3}$ and $\frac{2}{3}$ of the period $2\pi/\nu$.*

Proof. $1 + e^{i\theta} + e^{2i\theta} = e^{i\theta}(e^{-i\theta} + 1 + e^{i\theta})$. $\square$

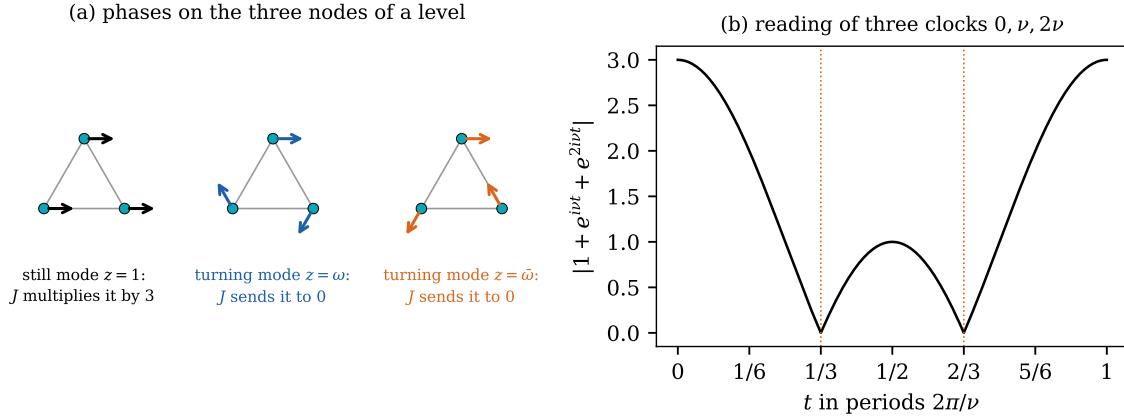


Figure 8. (a) The three phase patterns of a level: the still pattern, multiplied by 3, and the two turning patterns, erased. (b) The reading of three clocks at $0, \nu, 2\nu$ (Proposition 4), silent at $\frac{1}{3}$ and $\frac{2}{3}$ of the period.

9. Spectrum and Gap

Form: the graph and its matrices. Let P_L be the path graph on L vertices, with eigenvalues $\mu_k = 2 \cos \frac{k\pi}{L+1}$ [4, §1.4]. The *still sector* $\mathcal{S}$ of a strand is the space of vectors constant on each level, of dimension L , and the *turning sector* $\mathcal{T}$ is the space of vectors with zero sum on each level, of dimension $2L$.

- Theorem 6** (spectrum and gap). (i) *The adjacency matrix of a strand, $A(P_L) \otimes J$, has spectrum $\{3\mu_k\}_{k=1}^L$ and 0 with multiplicity $2L$; for $L = 6$ the nonzero values are $6 \cos \frac{k\pi}{7}$.*
- (ii) *$\mathcal{S}$ and $\mathcal{T}$ are invariant under the adjacency matrix, the degree matrix D and the Laplacian $\mathcal{L} = D - A$, and the adjacency matrix vanishes on $\mathcal{T}$.*
- (iii) *On $\mathcal{T}$, $\mathcal{L} = D$, with eigenvalues 3 on the end levels and 6 inside, for every L . On $\mathcal{S}$, $\mathcal{L}$ has eigenvalues $3(2 - 2 \cos \frac{k\pi}{L})$, $0 \leq k \leq L - 1$.*

Proof. The eigenvalues of a Kronecker product are the products of those of the factors [11, Theorem 4.2.12]; J annihilates zero-sum vectors and preserves constants, and the degree is constant on levels. On $\mathcal{S}$, in the basis $\frac{1}{\sqrt{3}}\mathbf{1}_\ell$, $\mathcal{L}$ is 3 times the Laplacian of P_L . $\square$

These facts follow at once from the Kronecker form. Their content is a split into two bands. The turning sector is a flat band: the adjacency matrix vanishes on it, so a turning pattern does not move from its level, and its Laplacian eigenvalues stay at 3 and 6 at every length. The still sector is a dispersing band, whose smallest positive eigenvalue $3(2 - 2 \cos \frac{\pi}{L}) \sim 3\pi^2/L^2$ is the algebraic connectivity of the strand in Fiedler's sense [5]. For $L = 2, 6, 12, 48, 96, 400$ the gaps are 3, 0.803848, 0.204445, 0.012846, 0.003212, 0.000185 (Figure 9b).

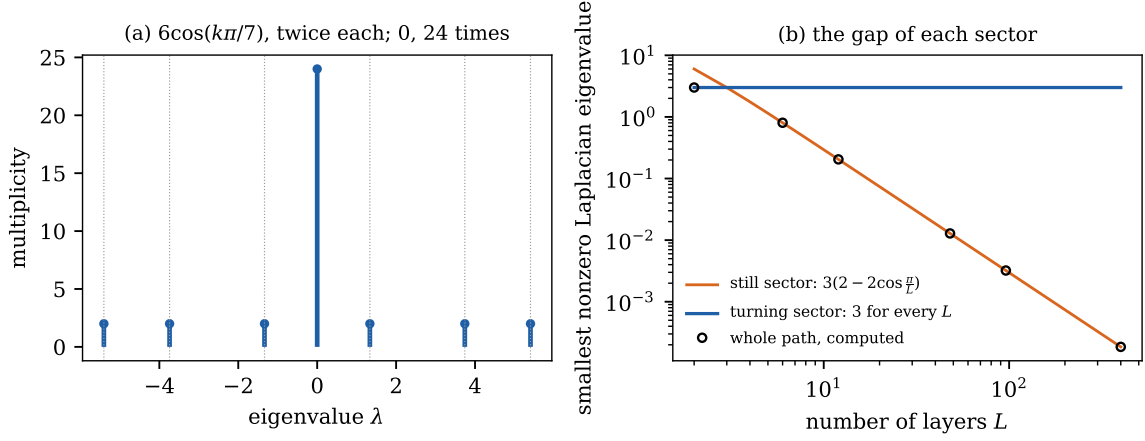


Figure 9. (a) Spectrum of the adjacency matrix of $\mathcal{A}_6$. (b) Smallest positive Laplacian eigenvalue of one strand against the number of layers: still sector, turning sector, and the whole strand.

10. The Ihara Zeta Function

Form: the graph and its closed walks. The Ihara zeta function of a finite graph [13] satisfies Bass's formula [3]

$$\zeta_G(u)^{-1} = (1 - u^2)^{|E| - |V|} \det(I - uA + u^2(D - I)). \quad (1)$$

A $(q + 1)$ -regular graph is Ramanujan when every eigenvalue $\lambda \neq \pm(q + 1)$ has $|\lambda| \leq 2\sqrt{q}$ [19]; equivalently, every nontrivial pole of ζ_G lies on $|u| = q^{-1/2}$ [25]. Writing $u = q^{-s}$, a factor $1 - \lambda u + qu^2$ of (1) vanishes exactly when $\lambda = q^s + q^{1-s}$, and $|u| = q^{-1/2}$ is the line $\Re s = \frac{1}{2}$.

- Theorem 7** (Ihara zeta). (i) $K_{3,3}$ is Ramanujan with $q = 2$: $\zeta_{K_{3,3}}(u)^{-1} = (1 - u)^4(1 + u)^4(1 - 2u)(1 + 2u)(1 + 2u^2)^4$. In the variable s its still eigenvalue $\lambda = 3$ sits at $s \in \{0, 1\}$, and its four turning eigenvalues $\lambda = 0$ at $s = \frac{1}{2} + \frac{(2k+1)\pi i}{2\log 2}$, $k \in \mathbb{Z}$, on the line $\Re s = \frac{1}{2}$.
- (ii) In a strand, with $q_\ell = \deg(\ell) - 1$ (2 at the ends, 5 inside), the turning sector contributes exactly the factor $(1 + 2u^2)^4(1 + 5u^2)^{2(L-2)}$ to (1): each of its poles lies on the circle $|u| = q_\ell^{-1/2}$ of its own level.

Proof. (i) $D - I = 2I$, A has eigenvalues $3, -3, 0, 0, 0, 0$ and $|E| - |V| = 3$; $2^s + 2^{1-s} = 3$ gives $2^s \in \{1, 2\}$, and $2^s + 2^{1-s} = 0$ gives $2^{2s} = -2$. (ii) The adjacency matrix and $D - I$ preserve both sectors, and on $\mathcal{T}$ the first vanishes while the second is q_ℓ on each level. $\square$

Since the nontrivial eigenvalues of $K_{3,3}$ are 0, the Ramanujan property is immediate. What the variable s adds is the placement. The pattern that a bridge keeps sits at $s = 1$, where the zeta function of the integers has its pole, and the patterns it erases sit on $\Re s = \frac{1}{2}$. The still sector of a whole strand is not regular, and its poles lie on no single circle; for $L = 6$ their moduli are 0.2306, 0.5319, 0.5538 and 1 (Figure 10). See also [15].

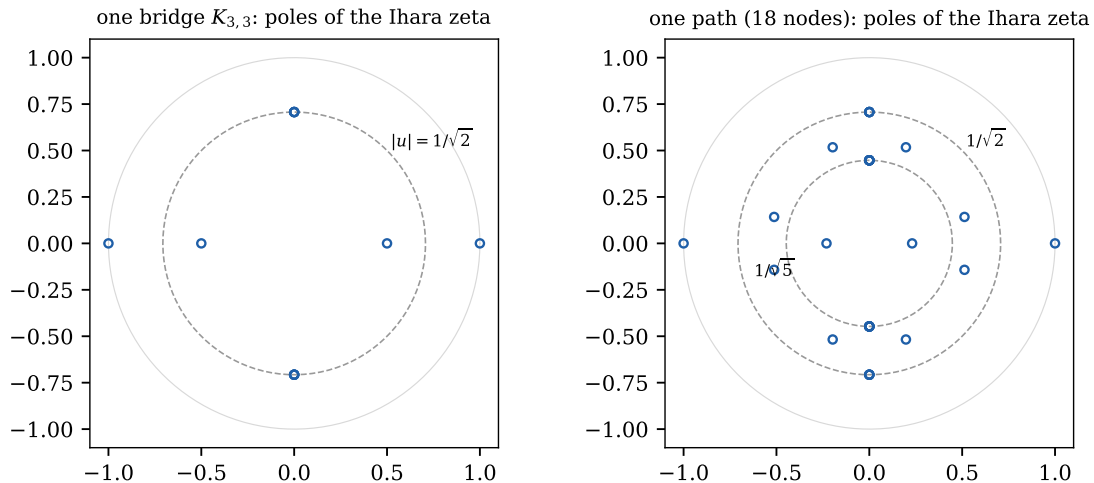


Figure 10. Poles of the Ihara zeta function. (a) One bridge: nontrivial poles on $|u| = 1/\sqrt{2}$. (b) One strand: the turning sector on $|u| = 1/\sqrt{2}$ (end levels) and $|u| = 1/\sqrt{5}$ (inner levels).

11. The Unrolled Strand

Form: the routes, unrolled. A route along a strand passes from a node of one level to a node of the next, keeping its angle or turning by $\pm 120^\circ$. We label these three steps by the digits 0 and ± 1 .

- Theorem 8** (the unrolled strand). (i) *The routes that start at a node v and descend one level at a time form the rooted tree T_3 in which every node has three children. The digits identify the endless routes with the 3-adic integers $\mathbb{Z}_3$, each written once as $\sum_{k \geq 0} d_k 3^k$ with $d_k \in \{-1, 0, 1\}$.*
- (ii) *At every node of T_3 the functions on its three children split as the values on a level do: into the constant, which a bridge multiplies by 3, and the two-dimensional space of zero-sum functions, which it erases. These zero-sum functions are the Haar wavelets of $\mathbb{Z}_3$ on that ball [16], so a bridge acts on $L^2(\mathbb{Z}_3)$ as one step of averaging over the balls.*
- (iii) *T_3 is the part below one vertex of the Bruhat–Tits tree of $\text{PGL}_2(\mathbb{Q}_3)$, in which every vertex has four neighbours [24]; this tree is the discrete hyperbolic plane on which p -adic holography is built [8]. The universal cover of a bridge is the tree in which every vertex has three neighbours, that of $\text{PGL}_2(\mathbb{Q}_2)$. A strand closed into a ring with an even number $L \geq 4$ of layers is 6-regular, and its universal cover is the tree of $\text{PGL}_2(\mathbb{Q}_5)$.*
- (iv) *The closed strand of (iii) is a Ramanujan graph exactly for $L \in \{4, 6, 8\}$.*

Proof. (i) The three neighbours of a node in the next level are reached by keeping the angle or turning by $\pm 120^\circ$, so every node of the route tree has three children. The digits $-1, 0, 1$ form a complete set of residues modulo 3, so every 3-adic integer has exactly one expansion in them [14, §4.1]. (ii) As in Proposition 3. The Haar basis of $L^2(\mathbb{Z}_3)$ consists of the constant together with, for every ball, the functions that are constant on its three sub-balls and sum to zero. (iii) The Bruhat–Tits tree of $\text{PGL}_2(\mathbb{Q}_p)$ is $(p+1)$ -regular, and the vertices below a given one, away from a fixed neighbour, form the tree in which every vertex has p children, whose ends are a ball of $\mathbb{P}^1(\mathbb{Q}_p)$ [24, Ch. II.1]. A connected $(q+1)$ -regular graph is covered by the $(q+1)$ -regular tree. In an even ring every node has three neighbours in each adjacent level. (iv) The closed strand is the cycle C_L blown up threefold, with eigenvalues $6 \cos \frac{2\pi k}{L}$ and 0. Its nontrivial eigenvalues satisfy $|\lambda| \leq 2\sqrt{5}$ exactly when $6 \cos \frac{2\pi}{L} \leq 2\sqrt{5}$, that is $L \leq 8$. $\square$

The bridge reads $L^2(\mathbb{Z}_3)$ one ball at a time. It passes the average of the three sub-balls and erases the wavelets, which are the turning patterns at that scale. The endless routes form a set without end that still has finite measure: $\mathbb{Z}_3$ is compact, of total Haar measure 1, and is made of infinite strings of the digits of the triad (Figure 11).

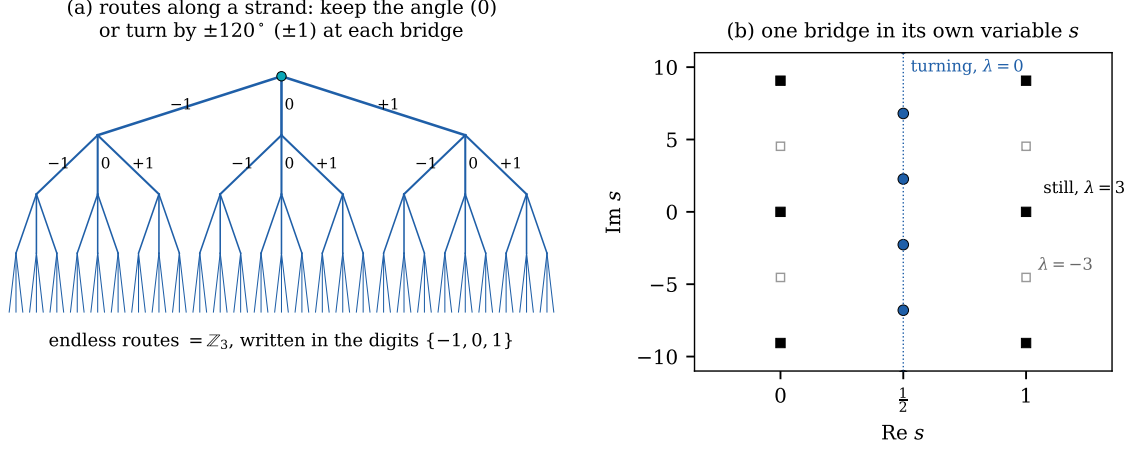


Figure 11. (a) The routes of a strand unrolled into the tree T_3 , with the digits of the first two steps; the ends are $\mathbb{Z}_3$. (b) One bridge in the variable s of Theorem 7: the still eigenvalue at $\Re s \in \{0, 1\}$, the turning eigenvalues on $\Re s = \frac{1}{2}$, repeated with period $2\pi / \log 2$ in $\Im s$.

12. The Axis as Time

Form: the strand as a chain of levels. Read along the axis, a strand is a chain of levels, and a bridge maps the state of one level to the next.

- Theorem 9** (transfer along the axis). (i) *The linear transfer of a bridge is J , with eigenvalue 3 on the still pattern and 0 on the turning patterns: the still pattern is the top state of the transfer, and a turning pattern does not survive one step.*
- (ii) *Put a triad spin $s_v \in \{-1, 0, 1\}$ on every node of a strand and couple the ends of every line by $e^{\kappa s_v s_w}$. The weight of a bridge depends on its two levels only through their sums $S, S' \in \{-3, \dots, 3\}$, and the transfer matrix is*

$$T_\kappa = \left(\sqrt{m(S)m(S')} e^{\kappa S S'} \right)_{S, S' = -3}^3, \quad (m(-3), \dots, m(3)) = (1, 3, 6, 7, 6, 3, 1),$$

the multiplicities being the coefficients of $(z^{-1} + 1 + z)^3$.

- (iii) *For every real κ the largest eigenvalue λ_0 of T_κ is simple and exceeds the modulus λ_1 of every other, so correlations along an endless strand decay as $(\lambda_1 / \lambda_0)^{|\ell - \ell'|}$, with mass $m(\kappa) = \log(\lambda_0 / \lambda_1) > 0$. At $\kappa = 0$, T_0 has rank one and $m(0) = \infty$.*

Proof. (i) Proposition 3. (ii) The nine lines of a bridge give $\sum s_v s_w = S_\ell S_{\ell+1}$, and $m(S)$ counts the configurations of a level with sum S . (iii) T_κ is a symmetric matrix with positive entries, so Perron's theorem applies [12, Ch. 8]; $T_0 = \sqrt{m} \sqrt{m}^T$. $\square$

The mass is $m(0.1) = 1.5879$, $m(\log \frac{3}{2}) = 0.083717$ and $m(\log 2) = 3.84 \times 10^{-4}$, and it decreases to 0 only as $\kappa \rightarrow \infty$. Read along the axis, then, the still pattern is not an excitation but the top state of the transfer. The turning patterns are confined to their level. The chain of triads has a gap at every coupling, in the endless limit as well, because each level is a triad and the transfer is a fixed finite positive matrix.

The same transfer gives the zeros in the fugacity $z^{\sum s_v}$. Lee and Yang proved that ferromagnetic systems of spins $\pm \frac{1}{2}$ have all their zeros in the fugacity on the unit circle [17]. Griffiths extended the theorem to higher spin [9], and Newman to every even single-site measure whose Laplace transform has only imaginary zeros [21]. The uniform triad $(1, 1, 1)$ has transform $1 + 2 \cosh h$, vanishing only at $h = \pm \frac{2\pi i}{3} + 2\pi i \mathbb{Z}$, so by Newman's theorem the zeros of a strand lie on the unit circle for every $\kappa \geq 0$.

Theorem 10 (the Lee–Yang circle). *Let $Z_A(z; \kappa) = \sum_{s_1, \dots, s_6} \prod_\ell m(S_\ell) z^{S_\ell} \prod_\ell e^{\kappa S_\ell S_{\ell+1}}$ for a strand of $\mathcal{A}_6$.*

- (i) For $\kappa = \log 2$ and $\kappa = \log \frac{3}{2}$ the 36 zeros of $z^{18}Z_A$ are simple and lie on $|z| = 1$.
- (ii) For $\kappa = -\log 2$ and $\kappa = -\log \frac{3}{2}$ no zero lies on $|z| = 1$.

Proof. Z_A is invariant under $z \mapsto z^{-1}$. Write $Z_A = c_0 + \sum_{k=1}^{18} c_k(z^k + z^{-k})$; then $Z_A(e^{i\theta}) = P(\cos \theta)$ with $P = c_0 + 2\sum_k c_k T_k$, T_k the Chebyshev polynomials, and P has rational coefficients since e^κ is rational. An exact Sturm count shows that P is squarefree with 18 roots in $(-1, 1)$ for $\kappa = \log 2, \log \frac{3}{2}$ and none in $[-1, 1]$ for $\kappa = -\log 2, -\log \frac{3}{2}$; also $Z_A(\pm 1) \neq 0$. Each root in $(-1, 1)$ gives the two zeros $e^{\pm i\theta}$. $\square$

At $\kappa = 0$ the zeros are those of $(z^{-1} + 1 + z)^{18}$, at one third and two thirds of a turn, and positive coupling keeps them on the circle. For $\kappa = 0.05, 0.1, 0.3, 0.5, 1$ they lie on the circle to within 10^{-51} ; for $\kappa = -0.1, -0.3, -1$ the largest moduli are 2.99, 16.3 and 1460 (Figure 12).

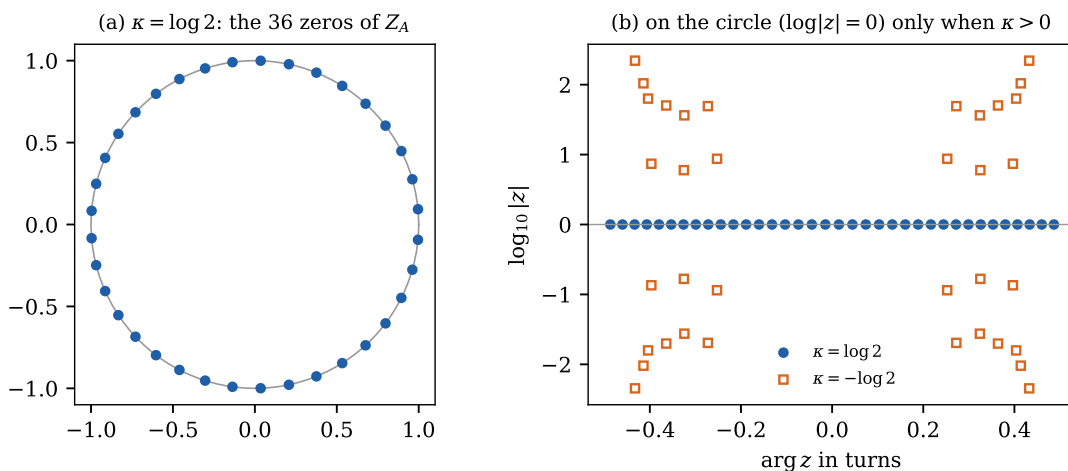


Figure 12. Lee-Yang zeros of one strand. (a) $\kappa = \log 2$: all 36 on the unit circle. (b) Argument and $\log_{10}$ modulus for $\kappa = \log 2$ and $\kappa = -\log 2$.

13. Without Nodes

Form: a smooth field. Let $\eta = (y - y_1)/Y$ count layers from the first, and put

$$\phi(\theta, \eta) = \cos(\pi\eta) \cos 3\left(\theta - \frac{\pi}{2}\right).$$

- Proposition 5** (the standing wave). (i) On every layer, $\phi = 1$ in the directions of the vertices and $\phi = -1$ in the directions of the midpoints.
- (ii) $\phi = \frac{1}{2} [\cos(3(\theta - \frac{\pi}{2}) - \pi\eta) + \cos(3(\theta - \frac{\pi}{2}) + \pi\eta)]$ is the mean of a left-handed and a right-handed screw, each turning the pattern through 60° per layer.
- (iii) ϕ vanishes on the horizons $\eta \in \mathbb{Z} + \frac{1}{2}$, the middle planes of the gaps.

Proof. (i) On layer i , $\cos(\pi\eta) = (-1)^{i-1}$, and $\cos 3(\theta - \frac{\pi}{2})$ equals 1 in the vertex directions of odd layers and -1 in those of even layers; the midpoint directions are opposite. (ii) is the identity $\cos a \cos b = \frac{1}{2} [\cos(a - b) + \cos(a + b)]$. (iii) $\cos \pi\eta = 0$ there. $\square$

The stack is what the two screws have in common. Each screw turns one way, and their mean turns neither way, flipping its sign from layer to layer and falling silent on the horizons (Figure 1h).

14. Methods

Computations are in Python with NumPy, SymPy for exact arithmetic and mpmath [20] for multiple precision; the scripts are provided with this article.

Graphs

The coordinates of Definition 1 with $R = 6$, $Y = \frac{9}{2}$; components, gradings, spectra, Laplacian sectors (an explicit orthonormal basis of $\mathcal{T}$, invariant to $5 \cdot 10^{-14}$ for $L \leq 400$), and the rings of Proposition 1 for $3 \leq L \leq 10$.

Symmetry

All compositions of a rotation about the axis by a multiple of 30° , or a reflection in a vertical plane at such an angle, with or without the reflection in the middle plane, tested on positions, colours and lines: exactly the 12 elements of Theorem 2 pass.

Crossings

Theorem 3 and Proposition 2 by solving the coefficient equations symbolically for all 81 index choices, and in exact rational arithmetic for $f = \frac{3}{10}, \frac{1}{2}, \frac{1}{3}$. All crossings of all pairs of the 90 segments by solving for the closest points of each pair (tolerance 10^{-9}), for triangle, pentagon and heptagon stacks: 30 meetings of the two strands and 15 self-crossings of each, at the heights and radii of Theorem 5.

Knots

Polygons projected in 12 random directions; crossings with over and under information give the Fox colouring matrix [7] and the Alexander polynomial by exact interpolation [2]; the code was checked on a trefoil, a figure-eight knot, a $(2, 5)$ torus knot and an unknot.

Trees, transfer and Lee–Yang

Balanced ternary digits of length 5 hit every residue modulo 3^5 exactly once. The eigenvalues of T_κ are computed in double precision. Z_A is computed exactly by the transfer, and the roots of P are counted by exact real-root isolation in SymPy; other κ use mpmath's polynomial root finder at 50 digits. Ihara poles are the reciprocals of the eigenvalues of $\begin{pmatrix} A & -(D-I) \\ I & 0 \end{pmatrix}$.

Smooth figures

Bands are parametrised as $(R_0 + v \cos \alpha(\phi))(\cos \phi, \sin \phi) + v \sin \alpha(\phi) e_z$, with α a smooth step from 0 to π centred at $\phi = \pi$ for the midpoint twist and $\alpha = \frac{3}{2}\phi$ for three half-twists.

15. Conclusions

One move builds the medial antiprism, a half-turn with a halving. Each step passes only a sum, and the answer to the opening question is exact. The still pattern survives every step: it is the top state of the transfer along the axis, and it sits at $s = 1$ in the variable of the unrolled tree. The patterns that turn by a third are erased at the first step: they form a flat band, they are confined to their level, and they sit on $\mathbb{R}s = \frac{1}{2}$. Where the erased parts meet is fixed by the geometry. A strand meets itself only at the thirds of each gap and meets the other strand only at the half. The meetings move to f and $1 - f$ when the midpoints are moved, and return to the half only at the midpoint. Each bridge holds a trefoil whose three crossings are undecided. An odd ring makes the two strands one, a path that returns only after two circuits.

The object was first drawn after a picture of anti-de Sitter space as a solid cylinder. Theorem 8 states its exact relation to that geometry. The routes of a strand unroll into the discrete hyperbolic plane of p -adic holography, whose ends are endless and of finite measure, and the axis carries a transfer with a gap. The paper leaves one question open: whether a metric arises from the structure in a limit, so that these discrete pieces join into a continuous geometry of anti-de Sitter type.

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